

		$\lambda \propto 1/\sqrt{E}$ The graph of λ vs $1/\sqrt{E}$ is a positive gradient straight line through the origin
29	D	Statement 1: False. The greater the binding energy <u>per nucleon</u> of a nucleus, the more stable it is. Statement 2: False. β decay often result in an emission of a daughter nucleus, an electron and a neutrino. It is unlikely that the 3 products will move on in opposite direction. Statement 3: False. A chemical reaction produces a new radioactive compound but the nuclei are unchanged. Thus, half-life is unchanged.
30	C	Initial $A_0 = \lambda N_0 = (\ln 2 / 4 \text{ years})N_0 = 0.1733 N_0$